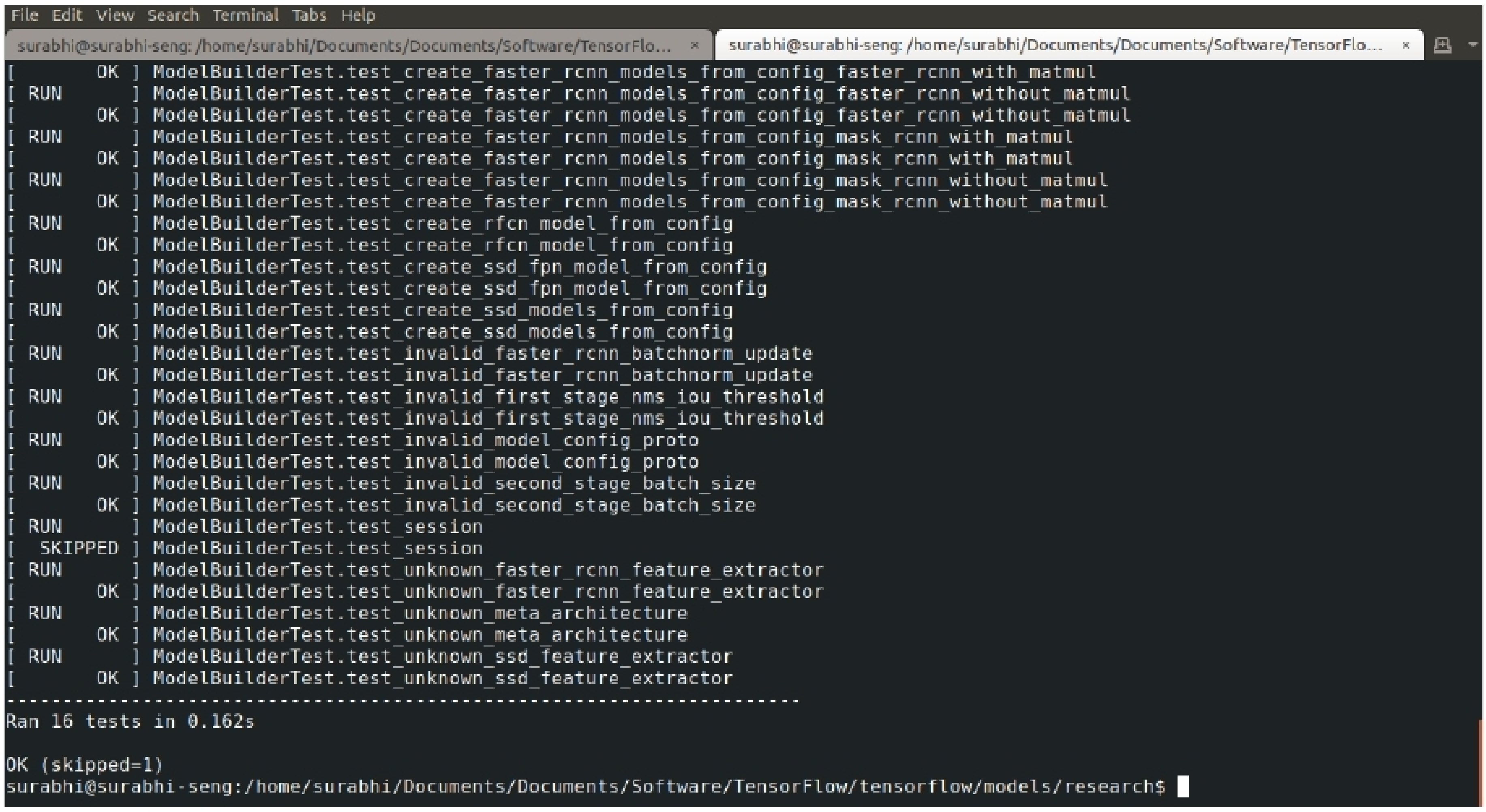

Figure 2: Testing the installation

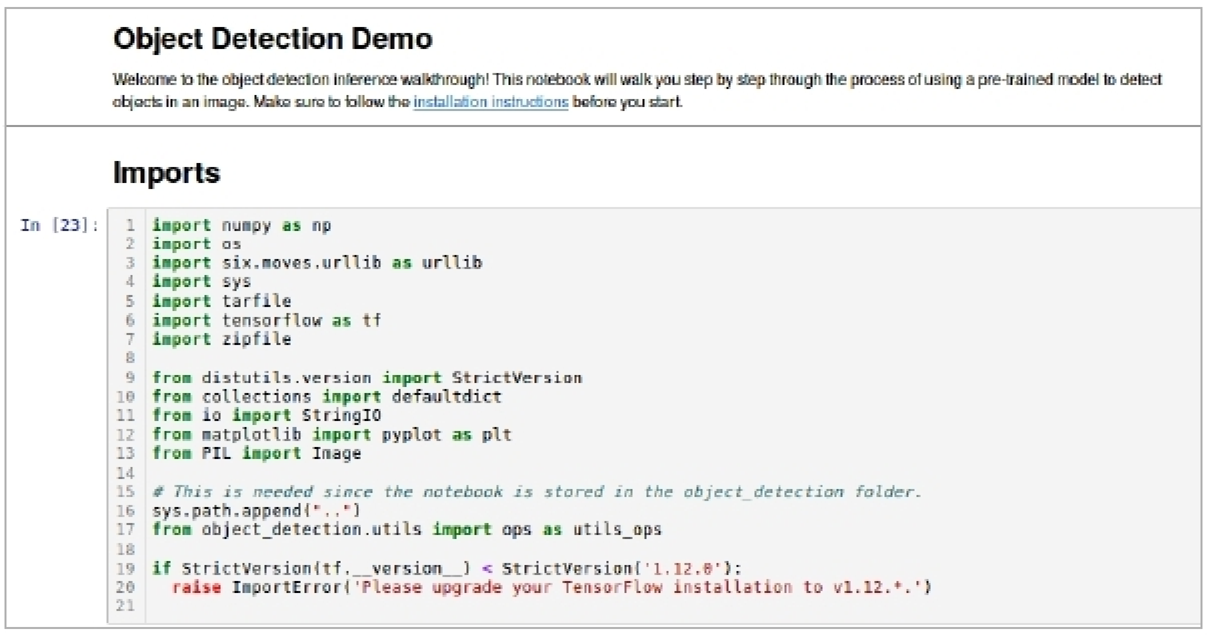

Figure 3: Code of object detection tutorial

models are composed of multiple processing layers. Deep learning methods have proved to be very effective in speech recognition, visual object recognition, object detection and many other domains.

In the subsequent section, I will discuss a very basic method of object identification using TensorFlow. This experiment has been carried on Ubuntu 18.04.3 with Python, TensorFlow and Protobuf 3.9.

The following steps can be used for object detection using TensorFlow. This will identify objects kept in the *test_images folder* of the TensorFlow directory.

- Set up the environment; install TensorFlow and the Tensor GPU using the *pip* command. Pip is a tool for installing and managing Python packages. Pip 19 or later is required for TensorFlow 2.0.

```
python -m pip install tensorflow
```

```
python -m pip install tensorflow-gpu
```

Install the required object detection libraries, as follows:

```
# static compiler
pip install --user Cython
# backport of the contextlib module to earlier Python versions.
pip install --user contextlib2
#python imaging library
pip install --user pillow
# For processing HTML and xml in python
pip install --user lxml
# open-source web application
pip install --user jupyter
#Plotting library for python
pip install --user matplotlib
```

Download Protocol Buffers (Protobuf). Run the following command from the *tensorflow/models/research/* folder:

```
protoc object_detection/protos/*.proto --python_out=.
```

Or you may compile each file in the *protos* folder, individually; for example: